PARTICLE SWARM OPTIMIZATION

CODE:

```
import numpy as np

def sphere_function(x):
    return np.sum(x**2)

num_particles = 30
num_dimensions = 2
num_iterations = 20
w = 0.7
c1 = 1.5
c2 = 1.5


positions = np.random.uniform(-10, 10, (num_particles, num_dimensions))
velocities = np.random.uniform(-1, 1, (num_particles, num_dimensions))

pbest_positions = positions.copy()
pbest_fitness = np.array([sphere_function(p) for p in positions])

gbest_index = np.argmin(pbest_fitness)
gbest_position = pbest_positions[gbest_index].copy()
gbest_fitness = pbest_fitness[gbest_index]

for iteration in range(num_iterations):
    for i in range(num_particles):

        r1, r2 = np.random.rand(), np.random.rand()
        velocities[i] = (w * velocities[i]
                         + c1 * r1 * (pbest_positions[i] - positions[i])
                         + c2 * r2 * (gbest_position - positions[i]))

        positions[i] += velocities[i]


        fitness = sphere_function(positions[i])


        if fitness < pbest_fitness[i]:
```

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```
            pbest_positions[i] = positions[i].copy()
            pbest_fitness[i] = fitness


            if fitness < gbest_fitness:
                gbest_position = positions[i].copy()
                gbest_fitness = fitness

    print(f"Iteration {iteration+1}/{num_iterations} - Best Fitness: 
{gbest_fitness:.6f}")

print("\nBest solution found:")
print("Position:", gbest_position)
print("Fitness:", gbest_fitness)
```

OUTPUT:

```
Iteration 1/20 - Best Fitness: 0.081095
Iteration 2/20 - Best Fitness: 0.081095
Iteration 3/20 - Best Fitness: 0.006725
Iteration 4/20 - Best Fitness: 0.003298
Iteration 5/20 - Best Fitness: 0.003298
Iteration 6/20 - Best Fitness: 0.003298
Iteration 7/20 - Best Fitness: 0.002956
Iteration 8/20 - Best Fitness: 0.002956
Iteration 9/20 - Best Fitness: 0.002956
Iteration 10/20 - Best Fitness: 0.002956
Iteration 11/20 - Best Fitness: 0.002956
Iteration 12/20 - Best Fitness: 0.001661
Iteration 13/20 - Best Fitness: 0.001066
Iteration 14/20 - Best Fitness: 0.001066
Iteration 15/20 - Best Fitness: 0.001066
Iteration 16/20 - Best Fitness: 0.000587
Iteration 17/20 - Best Fitness: 0.000587
Iteration 18/20 - Best Fitness: 0.000165
Iteration 19/20 - Best Fitness: 0.000095
Iteration 20/20 - Best Fitness: 0.000018

Best solution found:
Position: [ 0.00321441 -0.00268418]
Fitness: 1.7537291281392393e-05
```